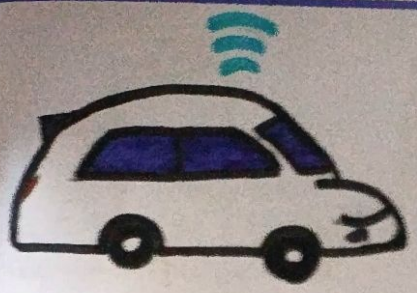
CO1 Assessment tool - 2

ITA0501 - Computer vision

Name : K. M. Santhiya

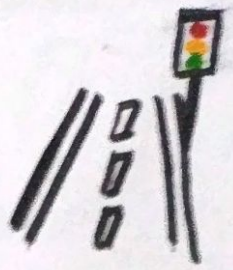
Reg No : 192371030

Date : 05.10.2026



Autonomous Vehicle

Concept Map



Overview of Computer Vision

- Enables machine to "see" and understand the visual world
- Uses cameras and Image processing to extract useful information
- Helps in object detection, recognition tracking and decision making

Features of Computer Vision

- Image capture and processing
- Feature extraction
- Object recognition
- Real time analysis
- Accuracy and reliability
- Works in various lighting conditions.

Levels of Computer Vision and Image processing.

Low Level → pre-processing (filtering, enhancement)

Mid Level → Feature extraction (edges, shapes, texture)

High Level → Object recognition (Classification, detection)

Digital Image Fundamentals

- Image = 2D array of pixels
- Each pixel has intensity / color value
- Represented in grayscale or RGB
- Uses spatial and intensity information.

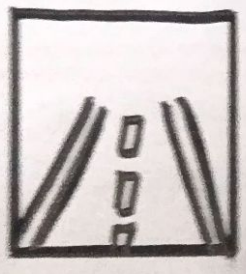


Image → pixel representation

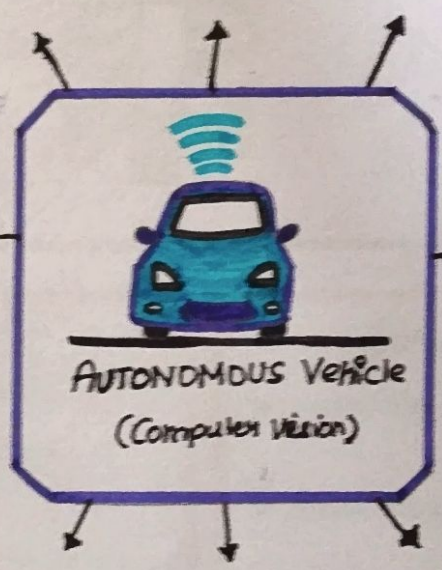


Image Sensing and Acquisition

- Cameras (front, rear, side)
- CMOS / CCD sensors
- Capture real-world light
- Convert to electrical signals



Sampling and Quantization

Sampling : Converts continuous image to discrete grid

Quantization: Converts intensity to digital value (eg: 8bit)

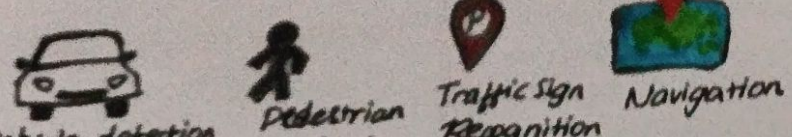
Continuous Image → Sampled & Quantized

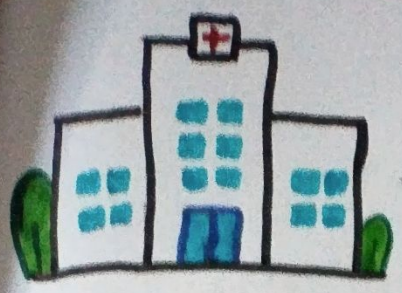
How Computer Vision Works in an Autonomous Car

1. Capture Images (cameras)
2. Process Image (filtering, enhancement)
3. Extract features (edge, shape, colour)
4. Recognize objects (AI/ML models)
5. Support navigation (decision making)

Applications

- Object detection (Vechicles, pedestrians, traffic signs)
- Lane detection
- Obstacle avoidance
- Traffic sign recognition
- Autonomous driving





Medical Imaging

Concept Map



Overview of Computer Vision

- Enable analysis and interpretation of medical Images
- Helps detects abnormalities and support diagnosis
- Uses image processing and AI techniques.

Features of Computer Vision

- Accurate and reliable analysis
- Feature extraction
- Object/anomaly detection
- Real time processing
- Handle different image types (X-ray, CT, MRI, etc.)

Levels of CV and Image Processing

- Low level → preprocessing (noise removal, enhance)
- Mid Level → Feature extraction (edges, texture, shapes)
- High level → Image interpretation (detection)

Digital Image Fundamentals

- Image = 2D array of pixels
- Grayscale or colored (R,G,B)
- Pixels intensity represents tissue density
- Higher bit depth = better detail



X-Ray image → Pixel representation

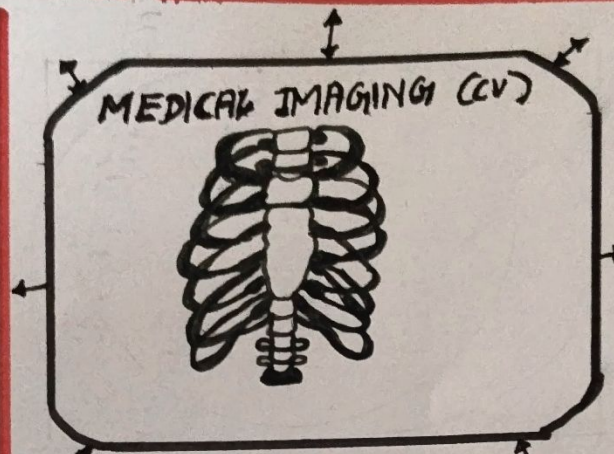


Image Sensing and Acquisition

- X-ray source (generates X-rays)
- Detector (flat panel / CCD)
- Captures X-ray pass-through
- Converts to electrical signal

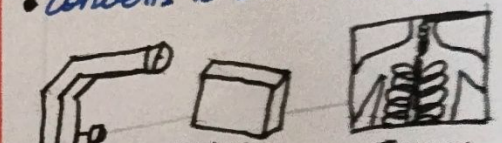
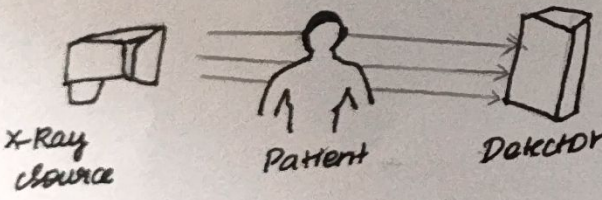


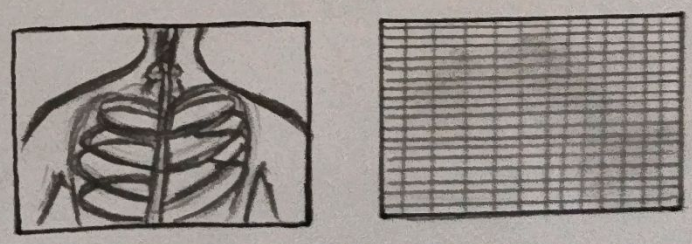
Image Formation Models

- X-ray attenuation (absorption)
- Different tissues absorb X-rays differently
- Mathematical model (Beer-Lambert law)



Sampling and Quantization

- Sampling: convert continuous X-ray to discrete grid
- Quantization: convert to digital intensity values (eg 16-bit)



How the Imaging System Works

1. Capture (X-ray passes through body)
2. Digitize (detect to digital signal)
3. Process (noise removal)
4. Analyse (feature extraction)
5. Support diagnosis (highlight abnormalities)

Applications in Medical Imaging

- Detecting fractures (X-ray)
- Identifying tumors (MRI/CT)
- Organ analysis (ultrasound)
- Disease screening (Chest X-ray)
- Treatment planning
- Monitoring patient progress.

X-ray → CT Scan → MRI → Ultra Sound.

Summary of computer vision Features in Medical Imaging.

- High accuracy and sensitivity
- Work with multiple modalities
- Real time and automated analysis
- Helps doctors make better decisions
- Improve patient care and outcomes